# Tessera Badge

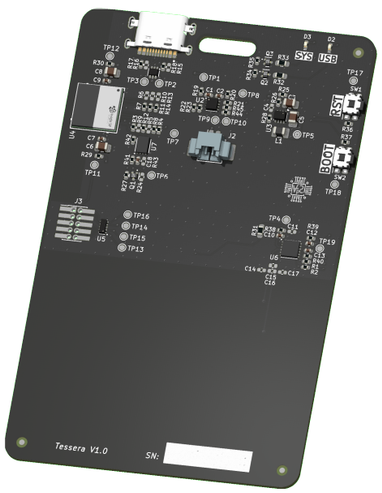
## Variant: PRELIMINARY

# 2026-10-11

# Rev + (Unreleased)


| Page | Index | Page | Index | Page | Index | Page | Index |
|---|---|---|---|---|---|---|---|
| 1 | Cover Page | 11 | ..... | 21 | ..... | 31 | ..... |
| 2 | Block Diagram | 12 | ..... | 22 | ..... | 32 | ..... |
| 3 | Project Architecture | 13 | ..... | 23 | ..... | 33 | ..... |
| 4 | USB | 14 | ..... | 24 | ..... | 34 | ..... |
| 5 | power | 15 | ..... | 25 | ..... | 35 | ..... |
| 6 | MCU | 16 | ..... | 26 | ..... | 36 | ..... |
| 7 | Leds | 17 | ..... | 27 | ..... | 37 | ..... |
| 8 | NFC | 18 | ..... | 28 | ..... | 38 | ..... |
| 9 | Secure element | 19 | ..... | 29 | ..... | 39 | ..... |
| 10 | Revision History | 20 | ..... | 30 | ..... | 40 | ..... |


## TOP VIEW




## BOTTOM VIEW




## DESIGN CONSIDERATIONS

DESIGN NOTE: Example text for informational design notes.

DESIGN NOTE: Example text for debug notes.

DESIGN NOTE: Example text for cautionary design notes.

**DESIGN NOTE: Example text for critical design notes.**

**LAYOUT NOTE: Example text for critical layout guidelines.**

## NOTES

Comment

Not fitted components are marked as X

**DRAFT - Very early stage of schematic, ignore details.**

**PRELIMINARY - Close to final schematic.**

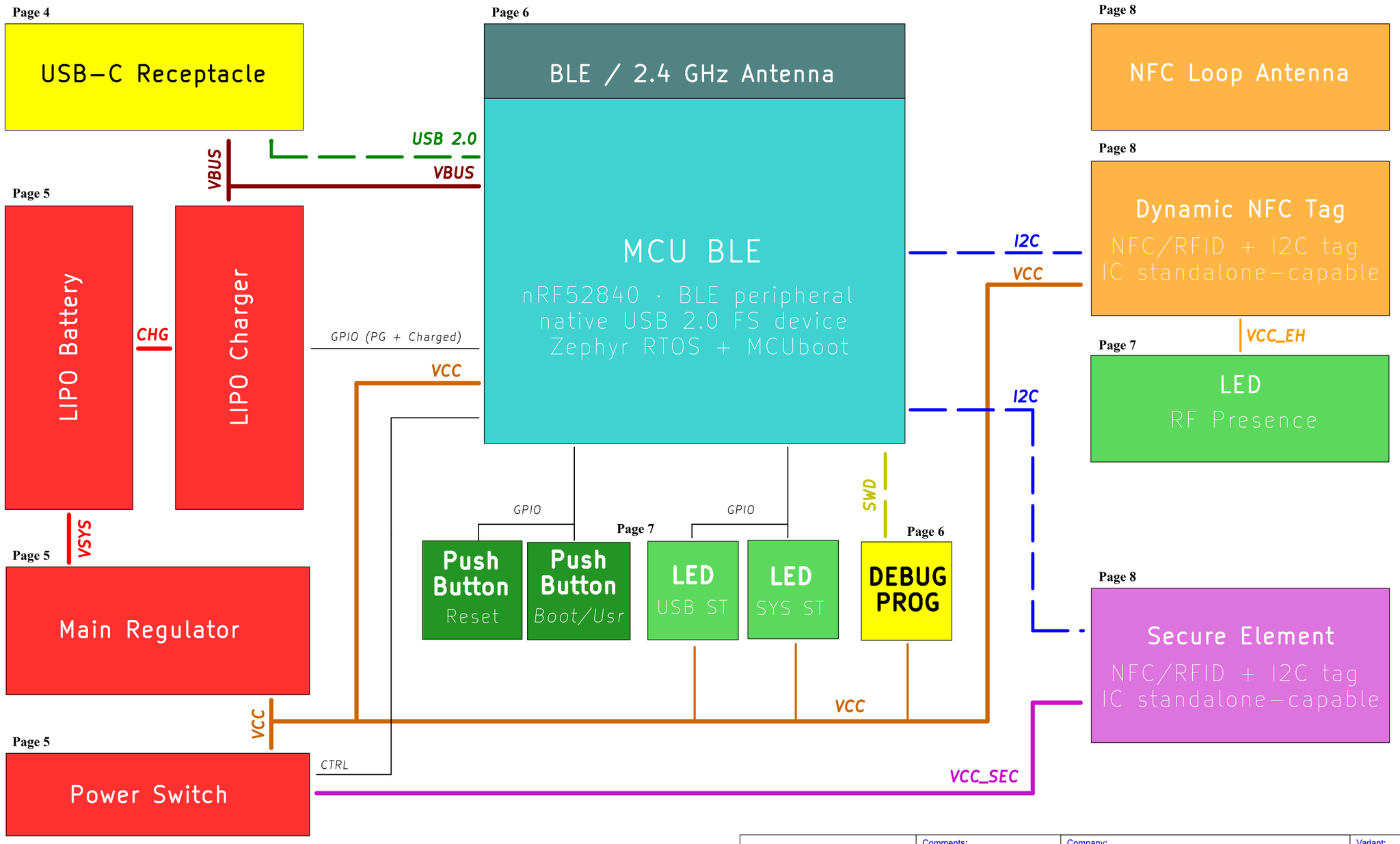
**CHECKED - There shouldn't be any mistakes. Contact the engineer if you find any.**

**RELEASED - A board with this schematic has been sent to production.**

**Date: 11-Oct-2026**

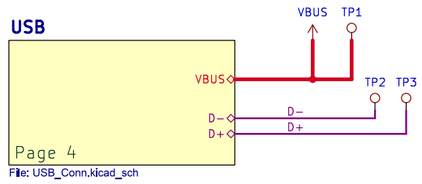
| | | | | |
|---|---|---|---|---|
| Comments: | Company: APS | | Variant: PRELIMINARY | Git Hash: 0342d82 |
| | Board Name: Tessera Badge | | Project Name: Tessera | |
| Sheet Title: Front Page | File Name: tessera.kicad_sch | Designer: Antonio Morales | Date: 2026-10-11 | Revision: + (Unreleased) |
| Sheet Path: / | | Reviewer: | Size: A3 | Sheet: 1 of 10 |

# [2] Block Diagram

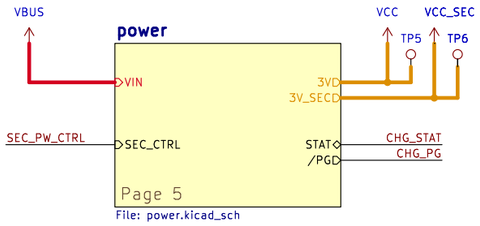



| | | | | |
|---|---|---|---|---|
| Comments: | Company: APS | | Variant: PRELIMINARY | Git Hash: 0342d82 |
| | Board Name: Tessera Badge | | Project Name: Tessera | |
| Sheet Title: Block Diagram | File Name: Block Diagram.kicad_sch | Designer: Antonio Morales | Date: 2026-10-11 | Revision: + (Unreleased) |
| Sheet Path: /Block Diagram/ | | Reviewer: | Size: A3 | Sheet: 2 of 10 |

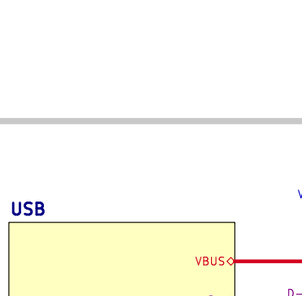
# [3] Project Architecture



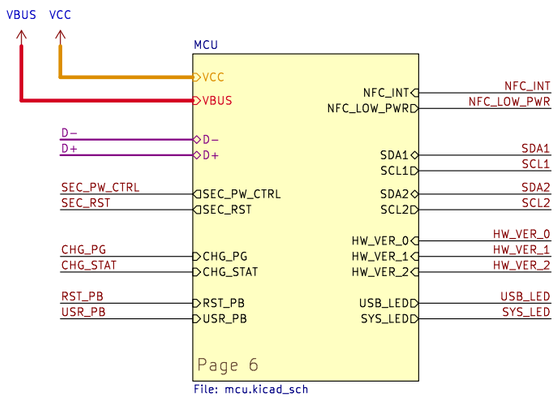

External power supply & USB connection



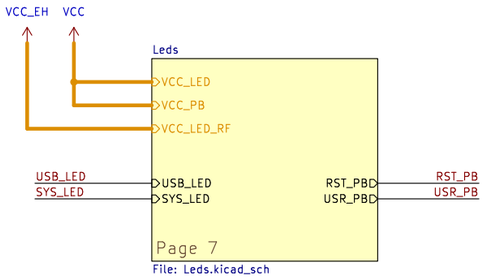

Batt charger & Main power supply



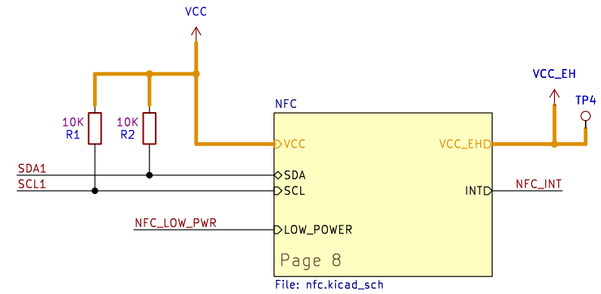

Board Hardware Version Selection 000b



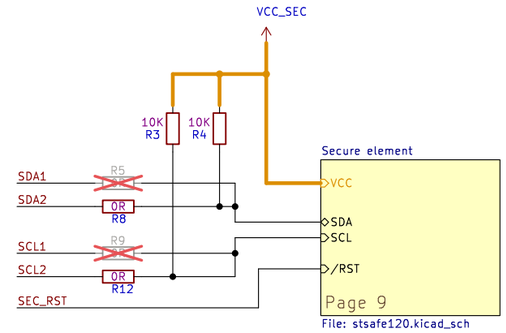

MCU & BLE Radio



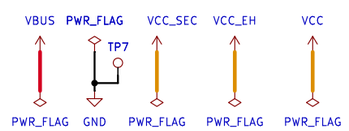

User IO: Leds & Push Button



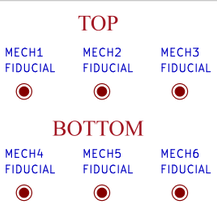

NFC Tag & Antenna




Secure Element Processor & Storage




ERC Power Flags




Fiducials

G1
WA_APS_Logo

G2
WA_tessera_Name

G3
WA_tessera_light_bulb

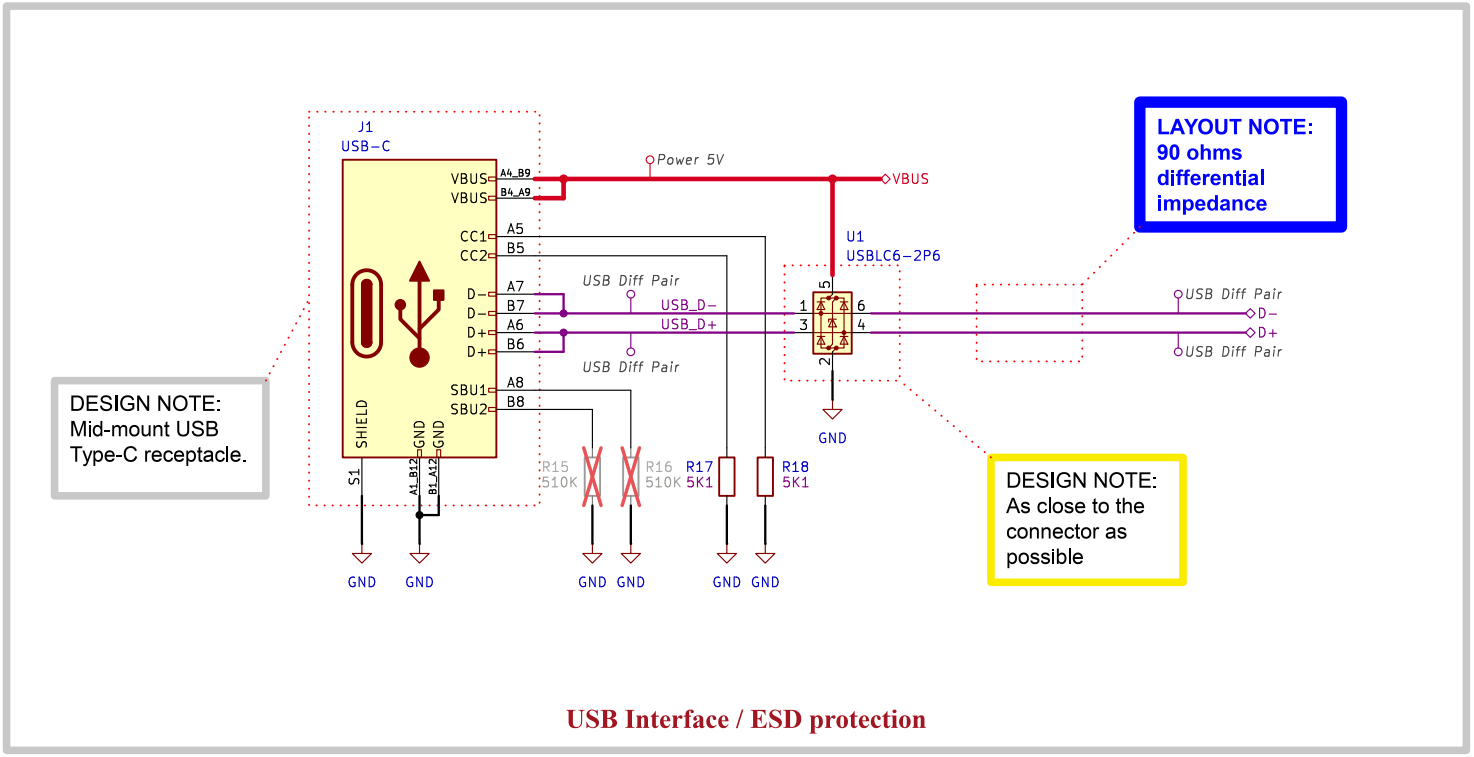
G4
WA_tessera_linkedin_qr

G5
WA_tessera_binary

Workart

| | | | | |
|---|---|---|---|---|
| Comments: | Company: APS | | Variant: PRELIMINARY | Git Hash: 0342d82 |
| | Board Name: Tessera Badge | | Project Name: Tessera | |
| Sheet Title: Project Architecture | File Name: Project Architecture.kicad_sch | Designer: Antonio Morales | Date: 2026-10-11 | Revision: + (Unreleased) |
| Sheet Path: /Project Architecture/ | | Reviewer: | Size: A3 | Sheet: 3 of 10 |

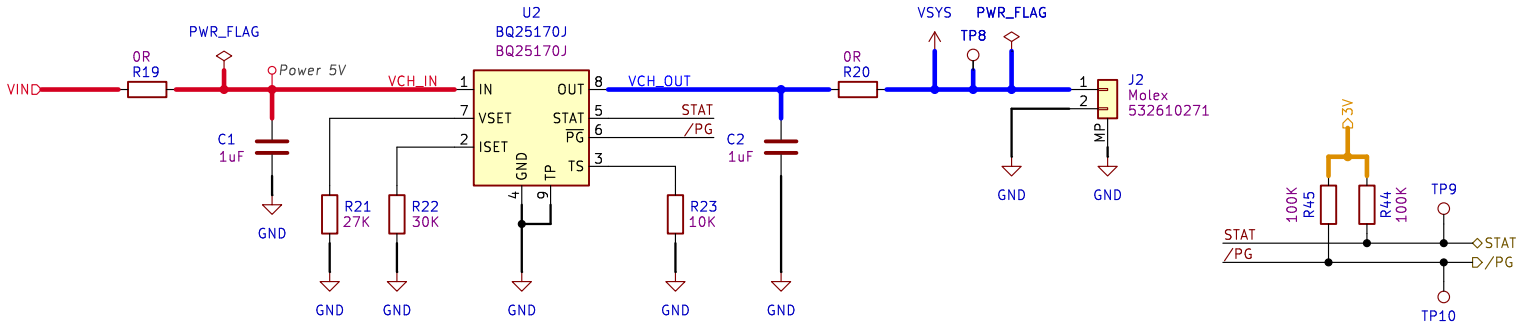
# [4] USB Connector



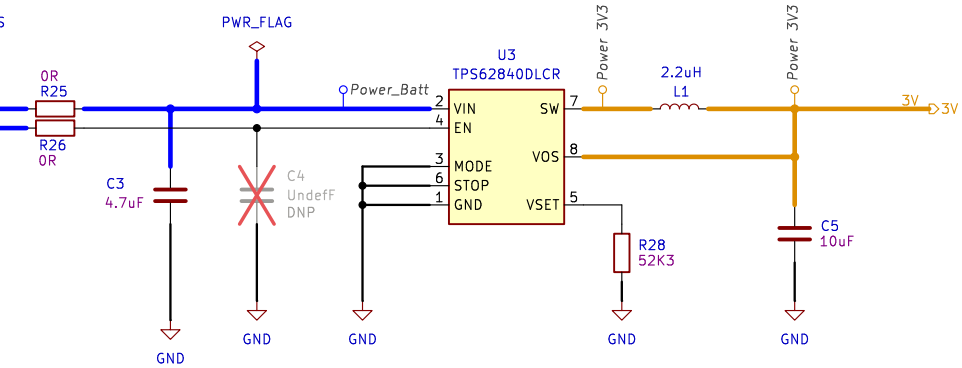

**USB Interface / ESD protection**

| | | | | |
|---|---|---|---|---|
| Comments: | Company: APS | | Variant: PRELIMINARY | Git Hash: 0342d82 |
| | Board Name: Tessera Badge | | Project Name: Tessera | |
| Sheet Title: USB Connector | File Name: USB_Conn.kicad_sch | Designer: Antonio Morales | Date: 2026-10-11 | Revision: + (Unreleased) |
| Sheet Path: /Project Architecture/USB/ | | Reviewer: | Size: A4 | Sheet: 4 of 10 |

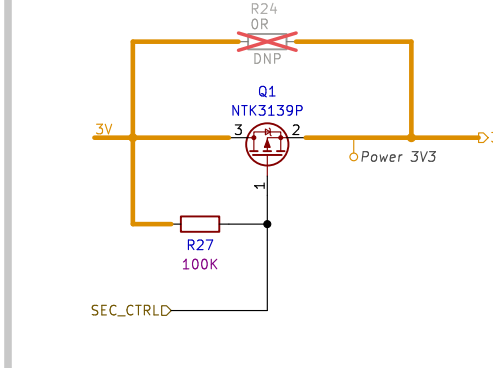
# [5] Power Supply




LIPO Battery Charger



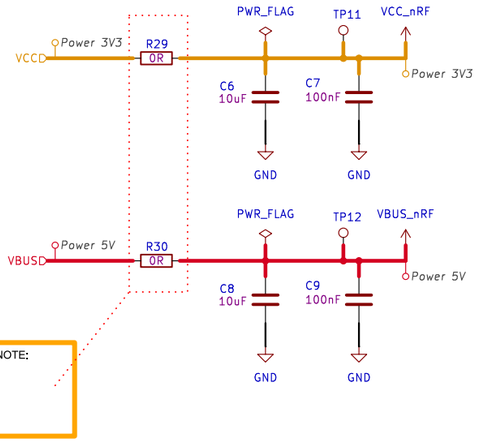

Main VCC DC/DC



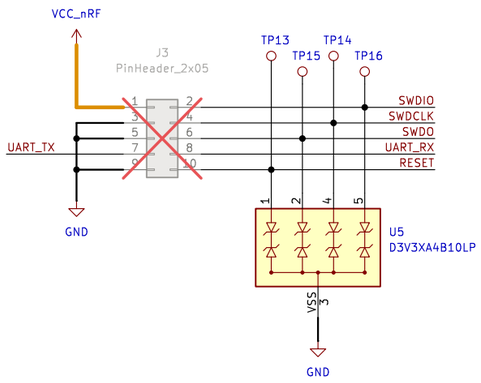

Secure element power switch

| | | | | |
|---|---|---|---|---|
| Comments: | Company: | | Variant: PRELIMINARY | Git Hash: 0342d82 |
| | Board Name: **Tessera Badge** | | Project Name: **Tessera** | |
| Sheet Title: Power Supply | File Name: power.kicad_sch | Designer: Antonio Morales | Date: 2026-10-11 | Revision: |
| Sheet Path: /Project Architecture/power/ | | Reviewer: | Size: **A4** | Sheet: **5** of **10** |

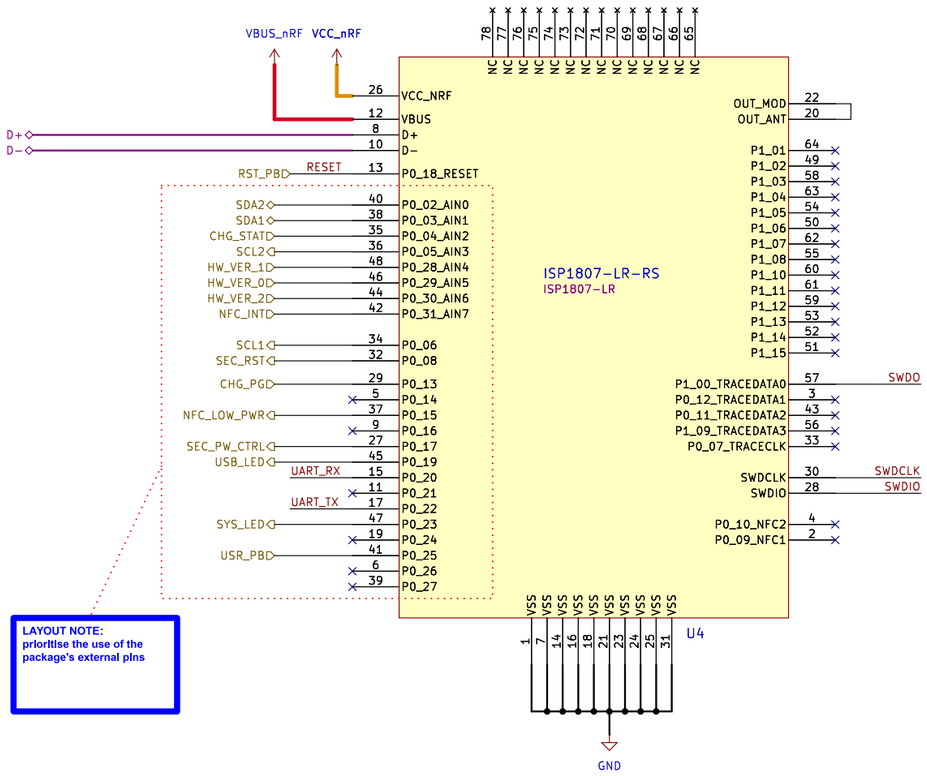
# [6] BLE MCU




MCU Power supply



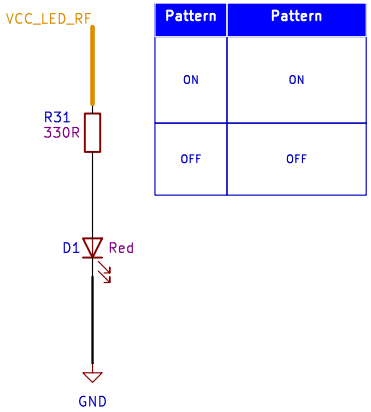

Debug & Program interface




Main MCU

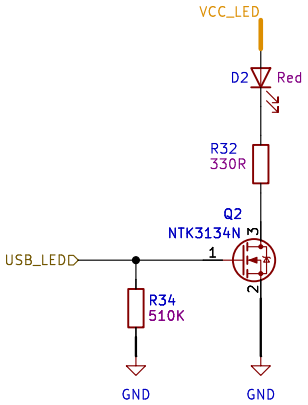
| | | | | |
|---|---|---|---|---|
| Comments: | Company: | | Variant: PRELIMINARY | Git Hash: 0342d82 |
| | Board Name: **Tessera Badge** | | Project Name: **Tessera** | |
| Sheet Title: BLE MCU | File Name: mcu.kicad_sch | Designer: Antonio Morales | Date: 2026-10-11 | Revision: |
| Sheet Path: /Project Architecture/MCU/ | | Reviewer: | Size: **A3** | Sheet: **6** of **10** |

# [7] User IO



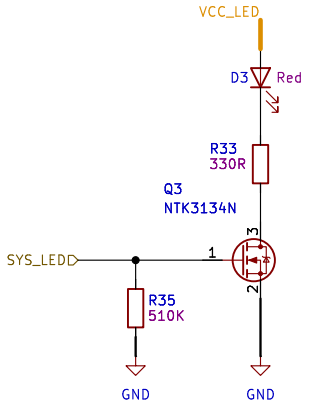

| Pattern | Pattern |
|---|---|
| ON | ON |
| OFF | OFF |

## NFC activity LED



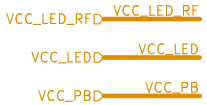

| Pattern | State |
|---|---|
| ON | Host has enumerated and mounted/opened the device (USB configured, crypto-token session open where applicable). Do not unplug. |
| OFF | Not on USB power, not yet enumerated, or session cleanly closed. Safe to unplug. |
| Blinking | |

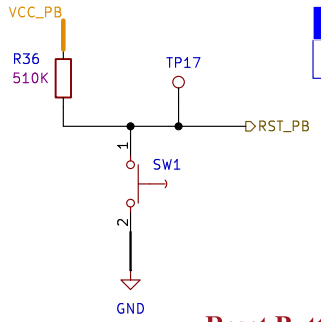
## USB Status LED




| Pattern | State |
|---|---|
| 500 ms ON, 1500 ms OFF, repeating | Bootloader / DFU active |
| 3× (50 ms ON, 200 ms OFF), one-shot | Application start |
| 5× (50 ms ON, 50 ms OFF), one-shot per event | NFC event while powered |
| 200 ms ON, 200 ms OFF, repeating | Crypto operation in progress |
| 20 ms ON every 2000 ms | BLE advertising |
| 2x 20 ms ON every 2000 ms | BLE connected |
| 1000 ms ON, 1000 ms OFF, repeating | USB charging in progress |
| Solid ON | USB charging complete |
| 100 ms ON, 100 ms OFF, continuous | Error |
| OFF | Idle |

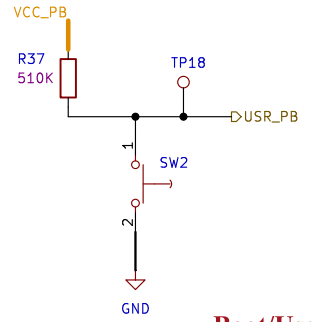
## System Status LED




| Action | Result |
|---|---|
| Push | Reset the device |

## Reset Button

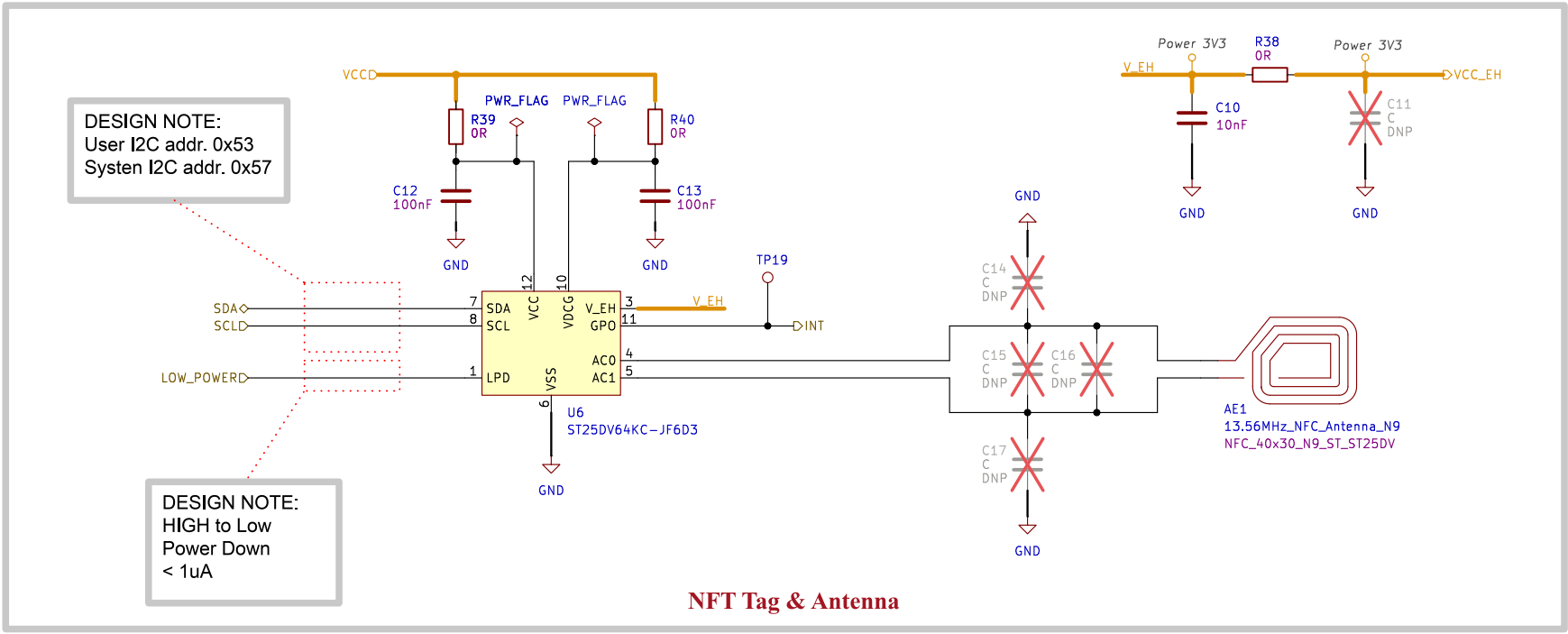



| Action | Result |
|---|---|
| Maintaid Pushed while booting | Stay in bootloader for DFU operation |
| Short push | Action 1 |
| Short push | Action 1 |
| Short push | Action 1 |

## Boot/User Button

| | | | | |
|---|---|---|---|---|
| Comments: | Company: | | Variant: PRELIMINARY | Git Hash: 0342d82 |
| | Board Name: **Tessera Badge** | | Project Name: **Tessera** | |
| Sheet Title: User IO | File Name: Leds.kicad_sch | Designer: Antonio Morales | Date: 2026-10-11 | Revision: |
| Sheet Path: /Project Architecture/Leds/ | | Reviewer: | Size: **A4** | Sheet: **7** of **10** |

# [8] NFC Interface

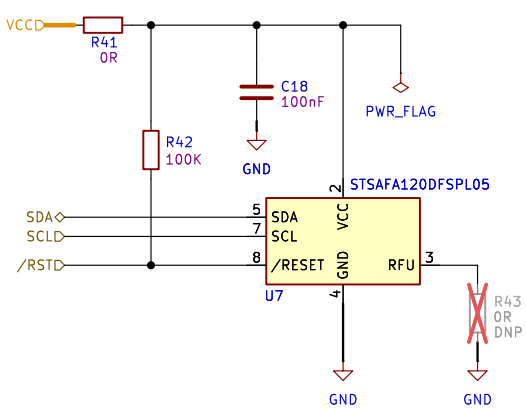



NFT Tag & Antenna

| Signal Name | Function | Direction |
|---|---|---|
| V_EH | Supply voltage for GPO driver | Power output |
| GPO | Interrupt output | Output |
| SDA | Serial data | I/O |
| SCL | Serial clock | Input |
| AC0, AC1 | Antenna coils | – |
| VCC | Supply voltage | Power |
| VSS | Ground | – |
| LPD | Low power down mode | Input |
| VDCG | Supply voltage for GPO driver | Power output |
| NC | Not connected | – |
| EP | Exposed pad | Must be left floating |

| | | | | |
|---|---|---|---|---|
| Comments: | Company: | | Variant: PRELIMINARY | Git Hash: 0342d82 |
| | Board Name: **Tessera Badge** | | Project Name: **Tessera** | |
| Sheet Title: NFC Interface | File Name: nfc.kicad_sch | Designer: Antonio Morales | Date: 2026-10-11 | Revision: |
| Sheet Path: /Project Architecture/NFC/ | | Reviewer: | Size: **A4** | Sheet: **8** of **10** |

# [9] Secure Element




NFT Tag & Antenna

| | | | | |
|---|---|---|---|---|
| Comments: | Company: | | Variant: PRELIMINARY | Git Hash: 0342d82 |
| | Board Name: **Tessera Badge** | | Project Name: **Tessera** | |
| Sheet Title: Secure Element | File Name: stsafe120.kicad_sch | Designer: Antonio Morales | Date: 2026-10-11 | Revision: |
| Sheet Path: /Project Architecture/Secure element/ | | Reviewer: | Size: **A4** | Sheet: **9** of **10** |

# [10] Revision History

| | | | | |
|---|---|---|---|---|
| Comments: | Company: APS | | Variant: PRELIMINARY | Git Hash: 0342d82 |
| | Board Name: **Tessera Badge** | | Project Name: **Tessera** | |
| Sheet Title: Revision History | File Name: Revision History.kicad_sch | Designer: Antonio Morales | Date: 2026-10-11 | Revision: + (Unreleased) |
| Sheet Path: /Revision History/ | | Reviewer: | Size: **A4** | Sheet: **10** of **10** |